

```

e2=e1*e1;
d1=n0+1;
d2=d1-2.0;
sum=0.0;
for (i=0;i<NMAX;i++,d1+=2.0,d2-=2.0,e1*=e2)
    sum += c[i]*(e1/d1+1.0/(d2*e1));
ans=0.5641895835*SIGN(exp(-xp*xp),x)*sum;    Constant is 1/√π.
}
return ans;
}

```

Other methods for computing Dawson's integral are also known [2,3].

CITED REFERENCES AND FURTHER READING:

- Rybicki, G.B. 1989, "Dawson's Integral and The Sampling Theorem," *Computers in Physics*, vol. 3, no. 2, pp. 85–87.[1]
 Cody, W.J., Pociorek, K.A., and Thatcher, H.C. 1970, "Chebyshev Approximations for Dawson's Integral," *Mathematics of Computation*, vol. 24, pp. 171–178.[2]
 McCabe, J.H. 1974, "A Continued Fraction Expansion, with a Truncation Error Estimate, for Dawson's Integral," *Mathematics of Computation*, vol. 28, pp. 811–816.[3]

6.10 Generalized Fermi-Dirac Integrals

The generalized Fermi-Dirac integral is defined as

$$F_k(\eta, \theta) = \int_0^\infty \frac{x^k (1 + \frac{1}{2}\theta x)^{1/2}}{e^{x-\eta} + 1} dx \quad (6.10.1)$$

It occurs, for example, in astrophysical applications with θ nonnegative and arbitrary η . In condensed matter physics one usually has the simpler case of $\theta = 0$ and omits the "generalized" from the name of the function. The important values of k are $-1/2$, $1/2$, $3/2$, and $5/2$, but we'll consider arbitrary values greater than -1 . Watch out for an alternative definition that multiplies the integral by $1/\Gamma(k+1)$.

For $\eta \ll -1$ and $\eta \gg 1$ there are useful series expansions for these functions (see, e.g., [1]). These give, for example,

$$\begin{aligned}
 F_{1/2}(\eta, \theta) &\rightarrow \frac{1}{\sqrt{2\theta}} e^\eta e^{1/\theta} K_1\left(\frac{1}{\theta}\right), & \eta \rightarrow -\infty \\
 F_{1/2}(\eta, \theta) &\rightarrow \frac{1}{2\sqrt{2}} \eta^{3/2} \frac{y \sqrt{1+y^2} - \sinh^{-1} y}{(\sqrt{1+y^2} - 1)^{3/2}}, & \eta \rightarrow \infty
 \end{aligned} \quad (6.10.2)$$

Here y is defined by

$$1 + y^2 = (1 + \eta\theta)^2 \quad (6.10.3)$$

It is the middle range of η values that is difficult to handle.

For $\theta = 0$, Macleod [2] has given Chebyshev expansions accurate to 10^{-16} for the four important k values, covering all η values. In this case, one need look no further for an algorithm. Goano [3] handles arbitrary k for $\theta = 0$. For nonzero θ ,

it is reasonable to compute the functions by direct integration, using variable transformation to get rapidly converging quadratures [4]. (Of course, this works also for $\theta = 0$, but is not as efficient.) The usual transformation $x = \exp(t - e^{-t})$ handles the singularity at $x = 0$ and the exponential fall off at large x (cf. equation 4.5.14). For $\eta \gtrsim 15$, it is better to split the integral into two regions, $[0, \eta]$ and $[\eta, \eta + 60]$. (The contribution beyond $\eta + 60$ is negligible.) Each of these integrals can then be done with the DE rule. Between 60 and 500 function evaluations give full double precision, larger η requiring more function evaluations. A more efficient strategy would replace the quadrature by a series expansion for large η .

In the implementation below, note how `operator()` is overloaded to define both a function of one variable (for `Trapzd`) and a function of two variables (for `DERule`). Note also the syntax

```
Trapzd<Fermi> s(*this,a,b);
```

for declaring a `Trapzd` object inside the `Fermi` object itself.

```
struct Fermi {
    Doub kk,etaa,thetaa;
    Doub operator() (const Doub t);
    Doub operator() (const Doub x, const Doub del);
    Doub val(const Doub k, const Doub eta, const Doub theta);
};
```

fermi.h

```
Doub Fermi::operator() (const Doub t) {
    Integrand for trapezoidal quadrature of generalized Fermi-Dirac integral with transformation
    x = exp(t - e^{-t}).
    Doub x;
    x=exp(t-exp(-t));
    return x*(1.0+exp(-t))*pow(x,kk)*sqrt(1.0+thetaa*0.5*x)/
        (exp(x-etaa)+1.0);
}
```

```
Doub Fermi::operator() (const Doub x, const Doub del) {
    Integrand for DE rule quadrature of generalized Fermi-Dirac integral.
    if (x < 1.0)
        return pow(del,kk)*sqrt(1.0+thetaa*0.5*x)/(exp(x-etaa)+1.0);
    else
        return pow(x,kk)*sqrt(1.0+thetaa*0.5*x)/(exp(x-etaa)+1.0);
}
```

```
Doub Fermi::val(const Doub k, const Doub eta, const Doub theta)
    Computes the generalized Fermi-Dirac integral  $F_k(\eta, \theta)$ , where  $k > -1$  and  $\theta \geq 0$ . The
    accuracy is approximately the square of the parameter EPS. NMAX limits the total number of
    quadrature steps.
    {
        const Doub EPS=3.0e-9;
        const Int NMAX=11;
        Doub a,aa,b,bb,hmax,olds,sum;
        kk=k;
        etaa=eta;
        thetaa=theta;
        if (eta <= 15.0) {
            a=-4.5;
            b=5.0;
            Trapzd<Fermi> s(*this,a,b);
            for (Int i=1;i<=NMAX;i++) {
                sum=s.next();
                if (i > 3)
                    if (abs(sum-olds) <= EPS*abs(olds))
```

Load the arguments into the member variables
for use in the function evaluations.

Set limits for $x = \exp(t - e^{-t})$ mapping.

Test for convergence.

```

        return sum;
    olds=sum;                                Save value for next convergence test.
}
}
else {
    a=0.0;                                    Set limits for DE rule.
    b=eta;
    aa=eta;
    bb=eta+60.0;
    hmax=4.3;                                Big enough to handle negative  $k$  or large  $\eta$ .
    DErule<Fermi> s(*this,a,b,hmax);
    DErule<Fermi> ss(*this,aa,bb,hmax);
    for (Int i=1;i<=NMAX;i++) {
        sum=s.next()+ss.next();
        if (i > 3)
            if (abs(sum-olds) <= EPS*abs(olds))
                return sum;
        olds=sum;
    }
}
throw("no convergence in fermi");
return 0.0;
}

```

You get values of the Fermi-Dirac functions by declaring a Fermi object:

```
Fermi ferm;
```

and then making repeated calls to the val function:

```
ans=ferm.val(k,eta,theta);
```

Other quadrature methods exist for these functions [5-7]. A reasonably efficient method [8] involves trapezoidal quadrature with “pole correction,” but it is restricted to $\theta \lesssim 0.2$. Generalized Bose-Einstein integrals can also be computed by the DE rule or the methods in these references.

CITED REFERENCES AND FURTHER READING:

- Cox, J.P., and Giuli, R.T. 1968, *Principles of Stellar Structure* (New York: Gordon and Breach), vol. II, §24.7.[1]
- Macleod, A.J. 1998, “Fermi-Dirac Functions of Order $-1/2$, $1/2$, $3/2$, $5/2$,” *ACM Transactions on Mathematical Software*, vol. 24, pp. 1–12. (Algorithm 779, available from netlib.)[2]
- Goano, M. 1995, “Computation of the Complete and Incomplete Fermi-Dirac Integral,” *ACM Transactions on Mathematical Software*, vol. 21, pp. 221–232. (Algorithm 745, available from netlib.)[3]
- Natarajan, A., and Kumar, N.M. 1993, “On the Numerical Evaluation of the Generalised Fermi-Dirac Integrals,” *Computer Physics Communications*, vol. 76, pp. 48–50.[4]
- Pichon, B. 1989, “Numerical Calculation of the Generalized Fermi-Dirac Integrals,” *Computer Physics Communications*, vol. 55, pp. 127–136.[5]
- Sagar, R.P. 1991, “A Gaussian Quadrature for the Calculation of Generalized Fermi-Dirac Integrals,” *Computer Physics Communications*, vol. 66, pp. 271–275.[6]
- Gautschi, W. 1992, “On the Computation of Generalized Fermi-Dirac and Bose-Einstein Integrals,” *Computer Physics Communications*, vol. 74, pp. 233–238.[7]
- Mohankumar, N., and Natarajan, A. 1996, “A Note on the Evaluation of the Generalized Fermi-Dirac Integral,” *Astrophysical Journal*, vol. 458, pp. 233–235.[8]